



Lab 06: Getting Started with Cloud KMS

Cloud Security Fundamentals | GSP079

Overview

Set up a secure Cloud Storage bucket, created a KeyRing and CryptoKey in Cloud KMS, encrypted financial data using the KMS API, uploaded encrypted files to Cloud Storage, and configured IAM permissions for key management.

Objectives

- Create a secure Cloud Storage bucket
- Enable Cloud KMS and create a KeyRing and CryptoKey
- Encrypt data using the KMS encrypt API endpoint
- Upload encrypted files to Cloud Storage
- Configure IAM permissions for KMS
- View Cloud Audit logs for KMS activity

Task 1 - Create a Cloud Storage Bucket

```
BUCKET_NAME="[PROJECT_ID]-kms_lab"  
gsutil mb gs://${BUCKET_NAME}
```

Task 2 - Review the Data

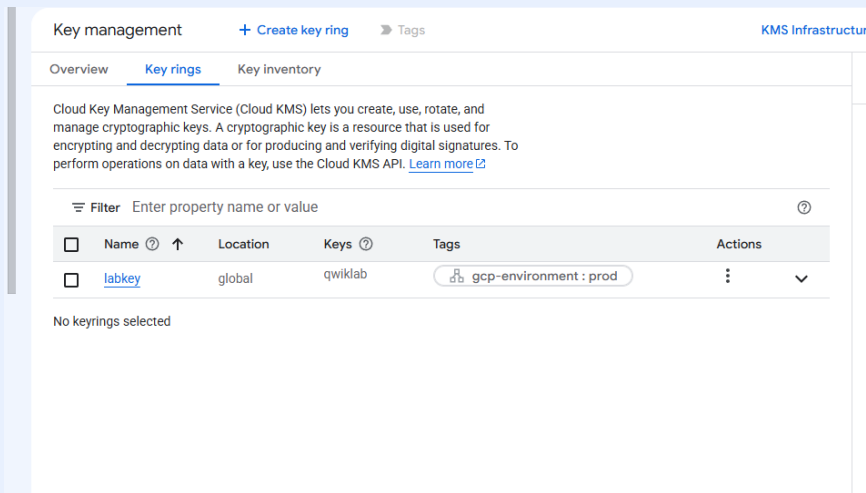
```
gsutil cp gs://${GOOGLE_CLOUD_PROJECT}-kms-lab-data/finance-dept/inbox/1.txt .  
tail 1.txt
```

Task 3 - Enable Cloud KMS

```
gcloud services enable cloudkms.googleapis.com
```

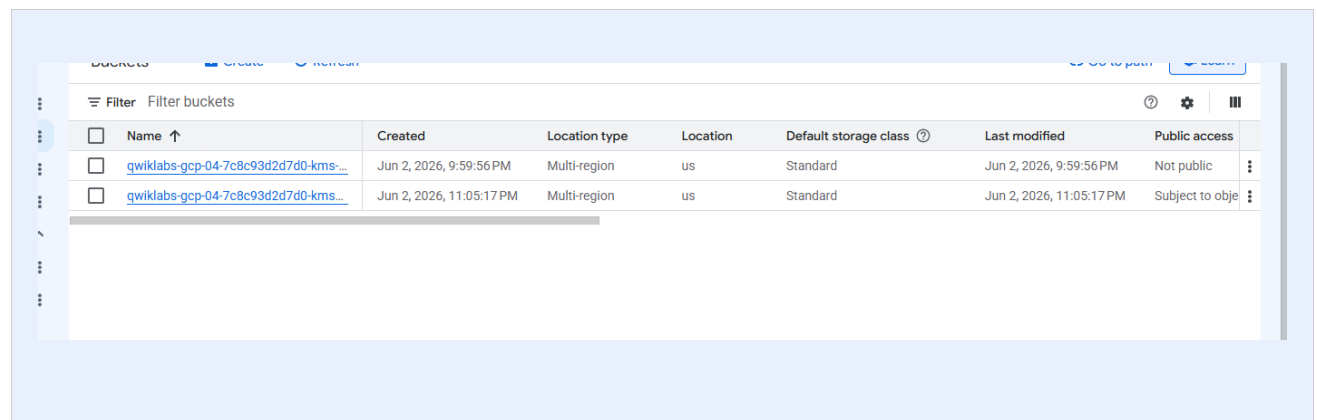
Task 4 - Create KeyRing and CryptoKey

```
KEYRING_NAME=labkey CRYPTOKEY_NAME=qwiklab  
gcloud kms keyrings create $KEYRING_NAME --location global  
gcloud kms keys create $CRYPTOKEY_NAME --location global --keyring $KEYRING_NAME --  
purpose encryption
```



Task 5 - Encrypt Data

```
PLAINTEXT=$(cat 1.txt | base64 -w0)
curl -v
"https://cloudkms.googleapis.com/v1/projects/$DEVSHIELD_PROJECT_ID/locations/global/
keyRings/$KEYRING_NAME/cryptoKeys/$CRYPTOKEY_NAME:encrypt" -d
{"plaintext":"$PLAINTEXT"}" -H "Authorization:Bearer $(gcloud auth
application-default print-access-token)" -H "Content-Type:application/json" | jq
.ciphertext -r > 1.encrypted
gsutil cp 1.encrypted gs://$BUCKET_NAME}
```



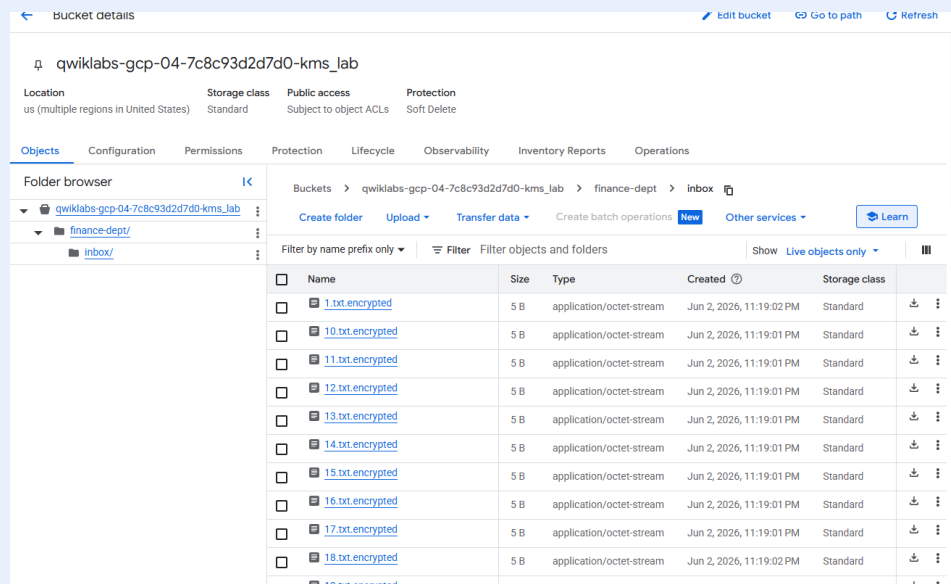
Task 6 - Configure IAM Permissions

```
USER_EMAIL=$(gcloud auth list --limit=1 2>/dev/null | grep '@' | awk '{print $2}')
gcloud kms keyrings add-iam-policy-binding $KEYRING_NAME --location global --member
user:$USER_EMAIL --role roles/cloudkms.admin
```

```
gcloud kms keyrings add-iam-policy-binding $KEYRING_NAME --location global --member user:$USER_EMAIL --role roles/cloudkms.cryptoKeyEncrypterDecrypter
```

Task 7 - Backup Data on Command Line

Encrypted all files in the finance-dept directory using a shell script and uploaded to Cloud Storage.

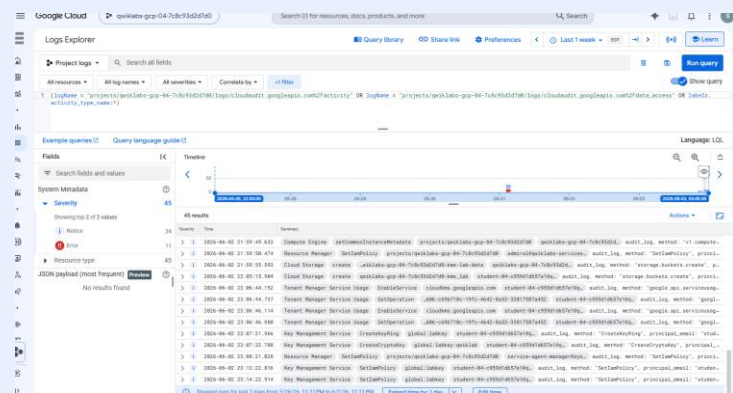


The screenshot shows the Google Cloud Storage interface for a bucket named 'qwiklabs-gcp-04-7c8c93d2d7d0-kms_lab'. The bucket is located in 'us (multiple regions in United States)', has a 'Standard' storage class, and is protected by 'Soft Delete'. The 'Objects' tab is selected, showing a folder browser on the left with 'finance-dept/' expanded to 'inbox/'. The main area displays a table of objects in the 'inbox' folder, all of which are encrypted .txt files. The table includes columns for Name, Size, Type, Created, and Storage class.

Name	Size	Type	Created	Storage class
1.txt.encrypted	5 B	application/octet-stream	Jun 2, 2026, 11:19:02 PM	Standard
10.txt.encrypted	5 B	application/octet-stream	Jun 2, 2026, 11:19:01 PM	Standard
11.txt.encrypted	5 B	application/octet-stream	Jun 2, 2026, 11:19:01 PM	Standard
12.txt.encrypted	5 B	application/octet-stream	Jun 2, 2026, 11:19:01 PM	Standard
13.txt.encrypted	5 B	application/octet-stream	Jun 2, 2026, 11:19:01 PM	Standard
14.txt.encrypted	5 B	application/octet-stream	Jun 2, 2026, 11:19:01 PM	Standard
15.txt.encrypted	5 B	application/octet-stream	Jun 2, 2026, 11:19:01 PM	Standard
16.txt.encrypted	5 B	application/octet-stream	Jun 2, 2026, 11:19:01 PM	Standard
17.txt.encrypted	5 B	application/octet-stream	Jun 2, 2026, 11:19:01 PM	Standard
18.txt.encrypted	5 B	application/octet-stream	Jun 2, 2026, 11:19:02 PM	Standard

Task 8 - View Cloud Audit Logs

Viewed Cloud KMS activity in Cloud Logging via Navigation Menu → Cloud Overview → Activity → View Log Explorer with Cloud KMS Key Ring as Resource Type.



What I Learned

- Cloud KMS manages encryption keys without exposing them to your application
- KeyRings group related CryptoKeys together by environment or purpose
- Data is base64 encoded before being sent to the KMS encrypt API
- KMS uses envelope encryption - data is encrypted with a data key, which is encrypted with a KMS key
- Cloud Audit logs track every key access and modification for compliance
- CryptoKeys and KeyRings cannot be deleted in Cloud KMS

Resources

GCP Cloud KMS Docs: <https://cloud.google.com/kms/docs>
 Lab ID: GSP079